Inputs

Electrode connector

Four differential filters

Common reference + bias

File: Inputs.kicad_sch

Frontend

ADS1299-4 acquisition

250 samples/s/channel

Reference / supply bypass

File: Frontend.kicad_sch

Controller

Cable to external ESP32

3.3 V logic / optional power

Wi-Fi HTTP + WebSocket target

File: Controller.kicad_sch

Power

Protected 2S battery input

3.3 V digital / 5 V analog

Removable enable shunt

File: Power.kicad_sch

USB

USB-C bench power only

Diode OR / analog 9 V boost

No charging / no USB data

File: USB.kicad_sch

4 channels + common reference + bias electrode; 250 samples/s/channel nominal.

Protected 2S battery, 6.4–8.4V; USB-C 5V bench adapter; external charging only.

External ESP32 via cable header; Wi-Fi firmware pending; USB-C is bench power only.

BENCH PROTOTYPE: validate using a simulator before any human connection.

Disconnect electrodes before charging, wired programming, or grounded instruments.

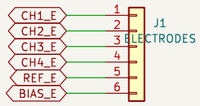
Sheet: /
File: EEG.kicad_sch

Title: Four-channel wireless EEG learning prototype

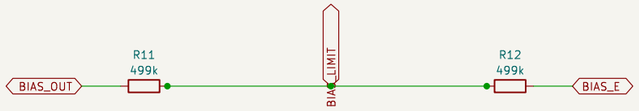
Size: A3	Date:	**Rev: B – bench prototype**
KiCad E.D.A. 9.0.9.1		Id: 1/6

ELECTRODES / INPUT PROTECTION

J1: channels 1–4, reference, bias



Bias electrode: two independent series resistors



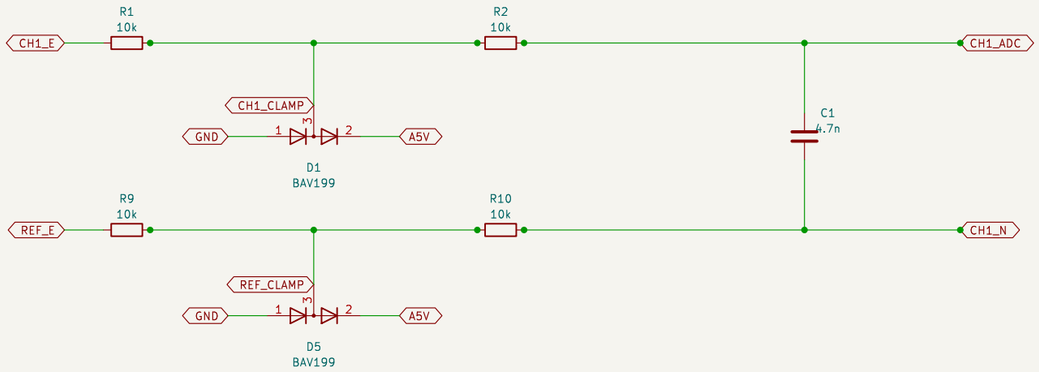
[I0] REF compares all channels; BIAS is a driven output.

J1 has no circuit-ground electrode. Trace CH1_E to CH1_ADC.

BIAS current path: R11 + R12 = 998 kOhm; bench validation pending.

CHANNEL 1 / DIFFERENTIAL FILTER

Positive electrode above; shared REF branch below



[I1] R1/R9 limit current toward the clamps; R2/R10 add isolation.

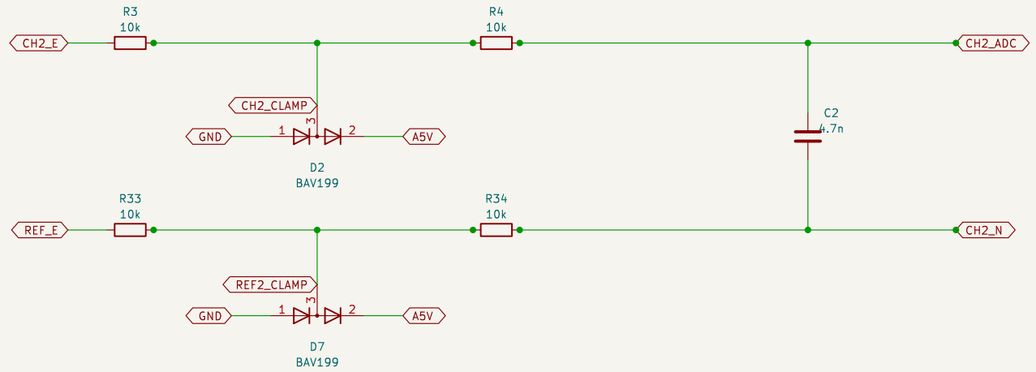
D1/D5: low-leakage rail clamps; leakage creates input offset.

C1 bridges P/N: 40 kOhm differential + 4.7 nF gives about 847 Hz.

This is high-frequency attenuation, not the final EEG bandpass.

CHANNEL 2 / DIFFERENTIAL FILTER

Positive electrode above; shared REF branch below



[I2] REF_E fans out into four separate protected branches.

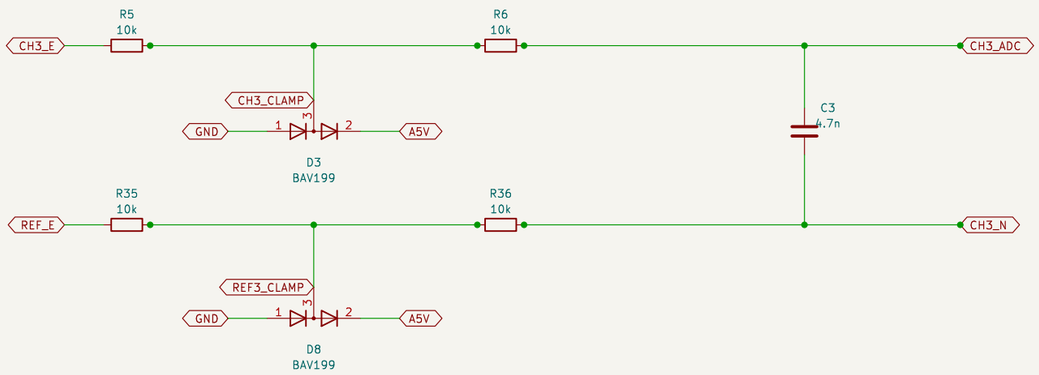
Each channel has its own negative input and differential capacitor.

A disturbance on the shared reference can affect several channels.

Channels 3/4 repeat this topology. Clamps do not certify safety.

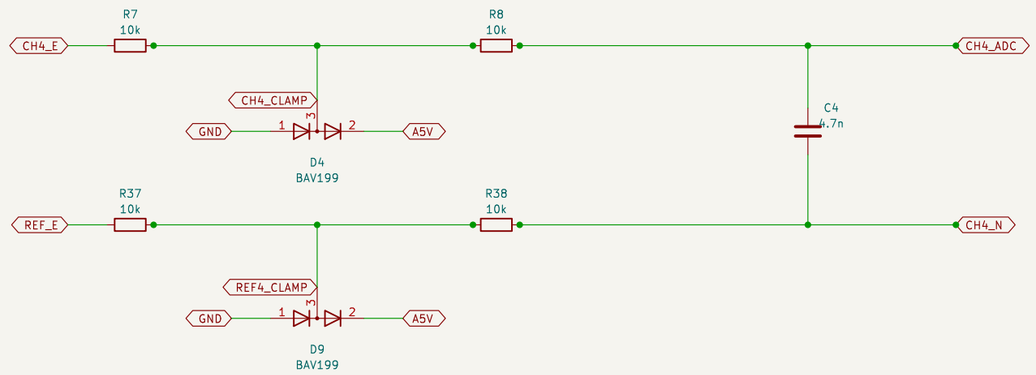
CHANNEL 3 / DIFFERENTIAL FILTER

Positive electrode above; shared REF branch below



CHANNEL 4 / DIFFERENTIAL FILTER

Positive electrode above; shared REF branch below



20 kOhm per input branch / 4.7 nF differential / rail clamps are not a certified patient interface

Sheet: /Inputs/
File: Inputs.kicad_sch

Title: EEG learning prototype – Inputs

Size: A3 | Date: 2026-10-13 | **Rev: B – bench prototype**

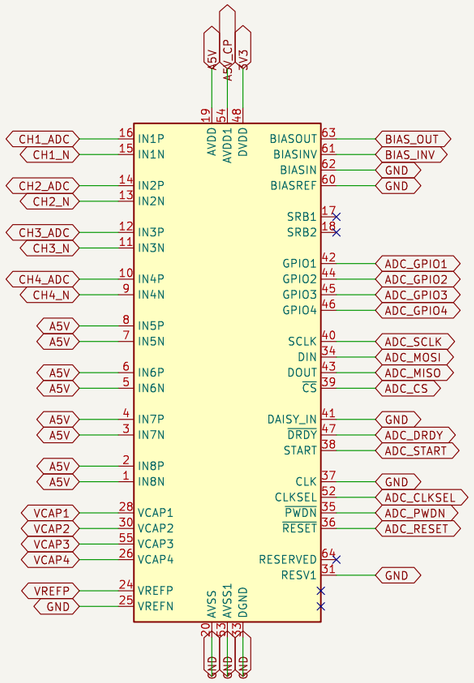
KiCad E.D.A. 9.0.9.1 | Id: 2/6

ADS1299-4 / ACQUISITION CORE

Four differential inputs from Inputs sheet; SRB1/SRB2 not used

[F1] U1 converts four differential channels simultaneously.
24-bit format does not imply 24-bit measurement accuracy.

U1
ADS1299-4

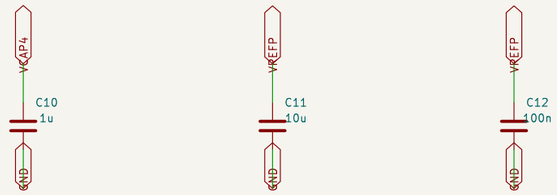


REFERENCE / VCAP BYPASS

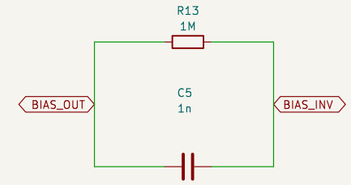
Follow TI placement guidance; C6 polarity matters



[F2] VCAP capacitors support internal circuits; not spare supplies.
C11/C12 steady the ADC conversion reference, not the REF electrode.
Use specified values and short local return paths; C6 is polarized.

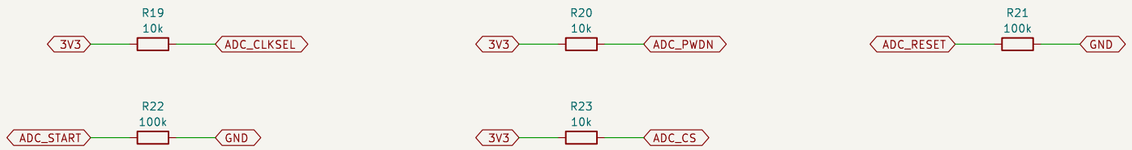


Bias amplifier feedback



[F3] R13 || C5 shapes bias feedback.
RC scale ~159 Hz; actual loop stability needs testing.
Use internal bias reference; enable after bench checks.

Hardware defaults

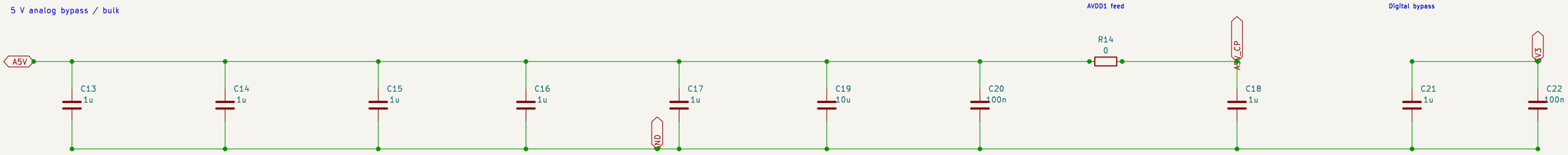


[F4] Pulls define startup: RESET/START low, CS high.
CLKSEL/PWDN high; firmware must configure the ADC.

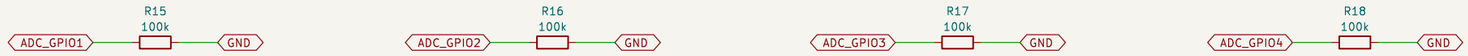
5 V analog bypass / bulk

AVDD1 feed

Digital bypass



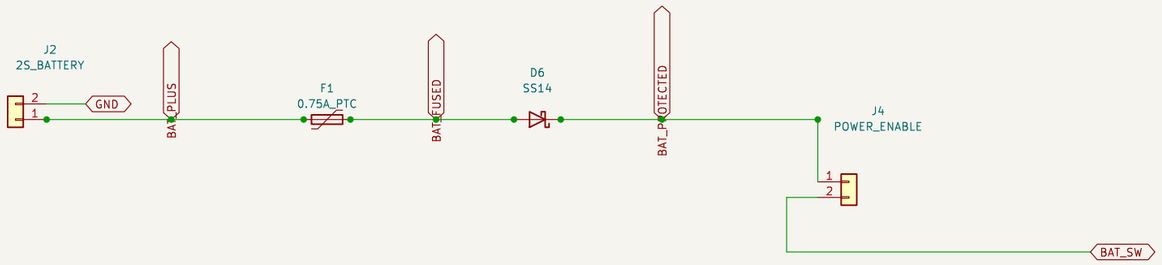
Unused GPIO pull-downs



[F5] Local bypass supplies transient current. R14 is a removable link, not a filter.

Sheet: /Frontend/	
File: Frontend.kicad_sch	
Title: EEG learning prototype — Frontend	
Size: A3	Date: 2026-10-13
KiCad E.D.A. 9.0.9.1	**Rev: B — bench prototype**
	Id: 3/6

BATTERY INPUT / DISCONNECT



[P1] F1: thermal overcurrent response, not a precise cutoff.

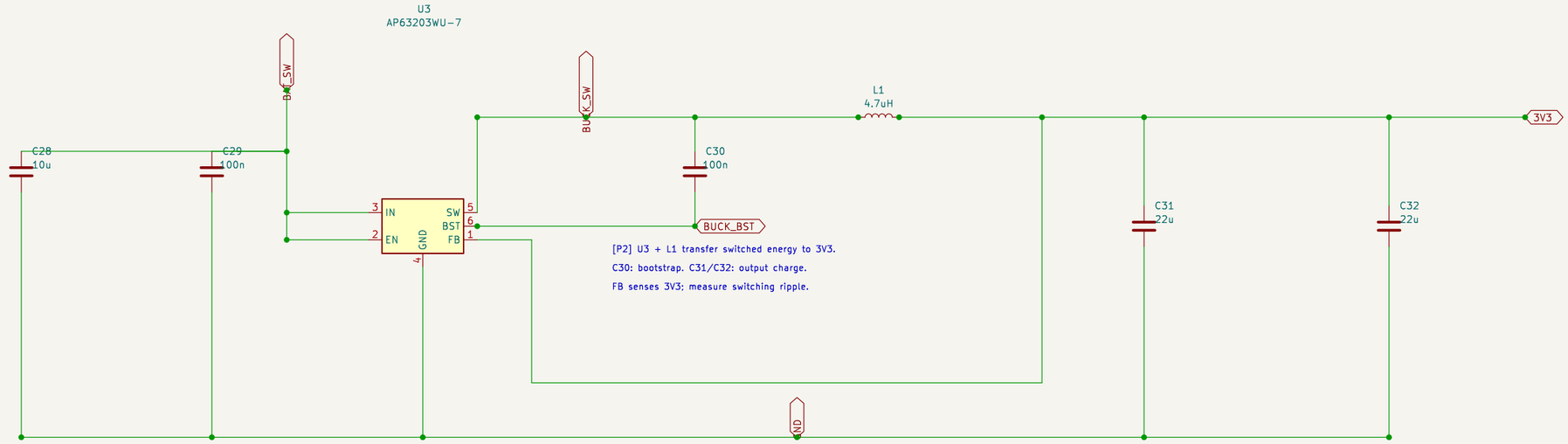
D6/D10 diode OR inputs; battery normally wins when both connected.

J4 disconnects BAT_SW; unplug the pack for master isolation.

2S battery / USB-C via diode OR / no charging

J4 shunt enables power; remove shunt or unplug battery for storage

3.3 V / DIGITAL SUPPLY



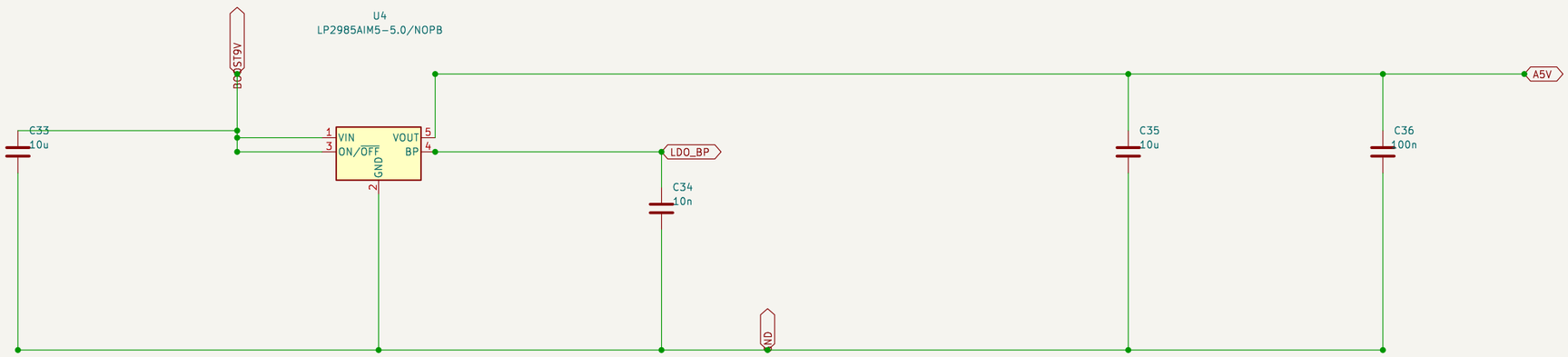
[P2] U3 + L1 transfer switched energy to 3V3.

C30: bootstrap. C31/C32: output charge.

FB senses 3V3; measure switching ripple.

[P3] Separate analog LDO avoids another local switch node; it is not galvanic isolation.

5 V / QUIET ANALOG SUPPLY

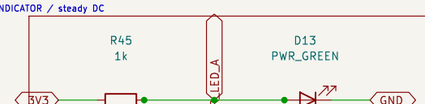


C34 bypasses the LDO noise-related node; C35 supports stability.

Check effective capacitance / ESR for the installed LP2985 revision.

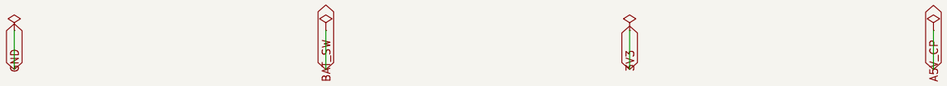
Boost gives analog headroom from USB; LDO loss ≈ (9.04−5) V × I.

3V3 POWER INDICATOR / steady DC



[P5] R45 = 1 kOhm; LED indicates 3V3 only, not analog readiness.

Power flags / ERC declarations



[P4] PWR_FLAG declares a driven rail for ERC; it adds no physical part.

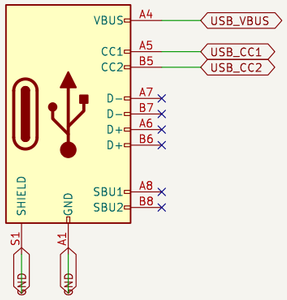
The pack needs independent protection. No onboard charger or hardware low-voltage cutoff.

Sheet: /Power/		
File: Power.kicad_sch		
Title: EEG learning prototype — Power		
Size: A3	Date: 2026-10-13	**Rev: B — bench prototype**
KiCad E.D.A. 9.0.9.1		Id: 4/6

USB-C BENCH INPUT / ANALOG BOOST

5 V supply only; no USB data, PD, or battery charging

J6
USB_C_BENCH_POWER



Independent CC pull-downs





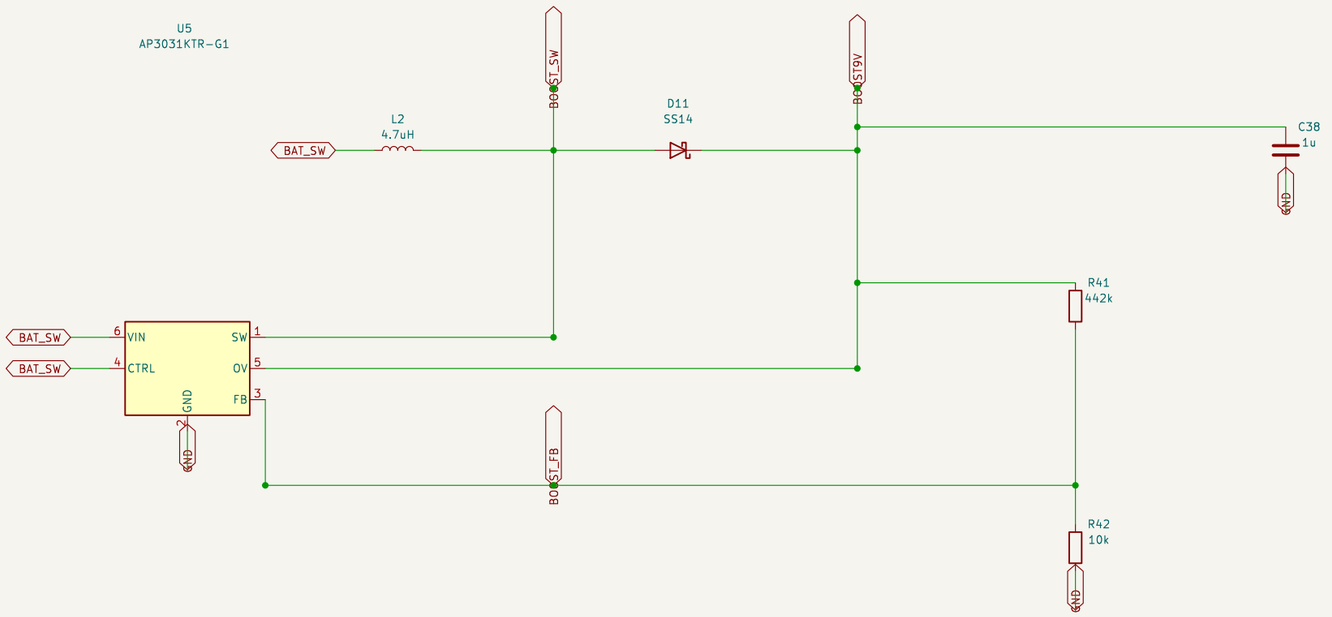
F2 + D10 feed shared protected input





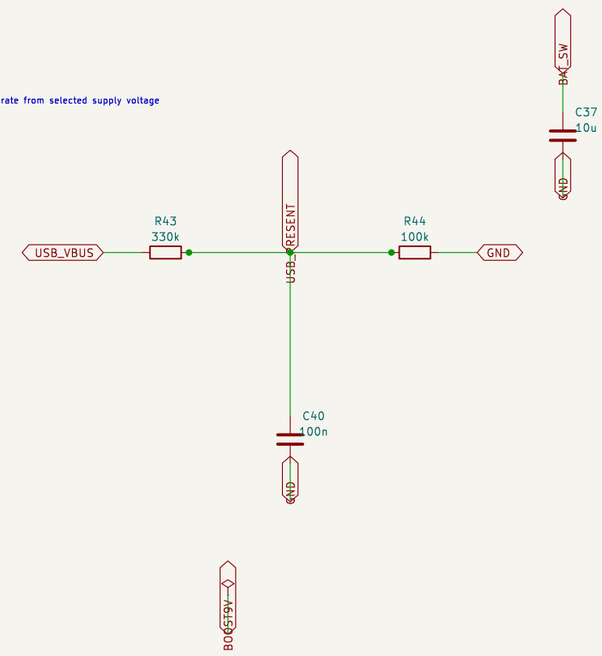
LOW-CURRENT ANALOG BOOST / 9.04 V NOMINAL

U5
AP3031KTR-G1



[U1] CC1/CC2 each need their own 5.1 kOhm sink resistor.
Use a 5 V bench adapter rated >=1 A; no PD negotiation.
No USB data is connected. Use the dev-board USB for programming.

USB presence monitor / separate from selected supply voltage



[U2] Vout = 0.2 V x (1 + 442k / 10k) = 9.04 V nominal.
U5 boosts BAT_SW only for the analog LDO; ESP32 uses the 3.3 V buck.
D6/D10 diode OR blocks backfeed; connecting USB still connects grounds.
Bench-only USB: disconnect electrodes before any wired connection.

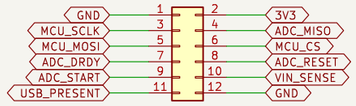
[U3] USB_PRESENT = VBUS / 4.3; analog presence indication, not USB negotiation.

Sheet: /USB/	
File: USB.kicad_sch	
Title: EEG learning prototype – USB	
Size: A3	Date: 2026-10-13
KiCad E.D.A. 9.0.9.1	**Rev: B – bench prototype**
	Id: 5/6

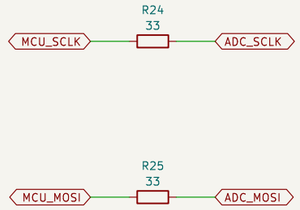
EXTERNAL ESP32 / CABLE INTERFACE

J5 is a signal cable header, not a plug-in dev-board socket

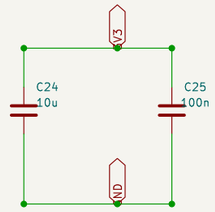
J5
EXTERNAL_ESP32_3V3



SPI series damping / 3.3 V logic



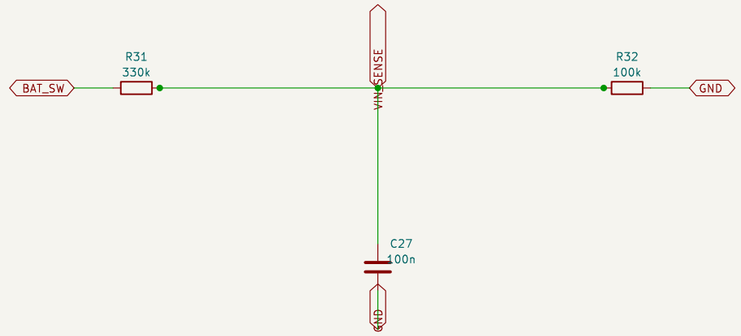
Local interface bypass



Automatic external-board power / disconnect power wire before its USB

[C1] J5 exposes SPI, DRDY, RESET, START, sense signals and ground.
Use short cables and 3.3 V GPIO. No onboard ESP32 or RF circuit.
Suggested S3 GPIO map is in README; verify the actual dev board.

Selected input voltage monitor / ratio 4.3



[C2] Wi-Fi target: HTTP UI + WebSocket frames; firmware pending.
Read on DRDY; use a bounded buffer so networking cannot block sampling.
Program through the external board USB, with electrodes disconnected.

[C3] J5 pin 2 is permanently supplied by the EEG 3.3 V rail.
Disconnect the J5 pin 2 power cable before its own USB.
Do not feed a dev-board 5 V pin from this 3.3 V output.
VIN_SENSE measures BAT_SW; USB_PRESENT identifies USB bench power.

[C4] R31/R32 ratio: 4.3. Verify ADC calibration and settling.
USB mode must not use the 6.4 V battery-mode cutoff threshold.

Sheet: /Controller/
File: Controller.kicad_sch

Title: EEG learning prototype — Controller

Size: A3	Date: 2026-10-13	**Rev: B — bench prototype**
KiCad E.D.A. 9.0.9.1		Id: 6/6